

Supplementary Materials for
LXR signaling controls homeostatic dendritic cell maturation

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Other Supplementary Material for this manuscript includes the following:

Data files S1 to S6
MDAR Reproducibility Checklist

SUPPLEMENTARY MATERIALS

Extended materials and methods

Immunogenic Maturation

To induce immunogenic maturation, mice were i.v. injected with 20-50 µg pIC (Sigma-Aldrich; P9582). 16h later, the mice were sacrificed.

CTLA4 blockade

To block the interaction of CD80 with CTLA4, mice were i.p. injected for 3 consecutive days with 100 µg anti-CTLA4 blocking antibody (Polpharma Biologics, clone 4F10) or isotype control (Thermo Fisher Scientific; 14-4888-85). Mice were sacrificed 24h after the last injection.

LNP formulation

LNPs were formulated according to the method described in (99). Aqueous solutions of pIC (Invivogen, tlr1-picw) were made by adding 750 µl of a pIC stock solution (1 mg/ml) to 9.25 ml 5 mM acetate buffer (pH 4) for pIC-LNPs. For eLNPs, aqueous solutions contained 10 ml 5 mM acetate buffer (pH 4). Ethanol solutions (5 ml) consisted of an ionizable lipid (ALC-0315; BroadPharm; BP-25498), 1,2-dimyristoyl-rac-glycero-3-methoxypoly (ethylene glycol) (DMG-PEG; PEG length, 2 kDa) (Avanti lipids; 880151P-1g), cholesterol (Sigma-Aldrich; C8667), dioleoylphosphatidylethanolamine (DOPE; Avanti lipids; 850725P-25mg) and DOPE-Cy5 (Avanti lipids; 810335C-1mg) or DiD (Thermo Fisher Scientific; D307) (see **Table S1** for LNP composition). LNPs were fabricated by solvent displacement. Hereto, an ethanolic solution containing all lipids was added to an aqueous solution containing pIC (or blank) under vigorous mixing on a vortex mixer. To remove ethanol, the formed LNP suspensions were dialysed overnight against phosphate buffered saline (PBS) using Slide-A-Lyzer(r) dialysis cassettes (cut-off 3.5 kDa) (Thermo Fisher Scientific; 66107). Subsequently, the dialysed LNP suspensions were concentrated 10X using Amicon Ultra 10K Centrifugal Filters (Millipore; UFC910024).

Diphtheria Toxin injections

CD45.1 mice, reconstituted with the BM of *Xcr1*-cre *Rosa26*-LSL-iDTR mice, were i.p. injected with 500 ng Diphtheria toxin (DT; Sigma-Aldrich; D0564) at different time points (0h, 12h, 24h, 36h, 48h, 72h, 96h, 108h, 120h and 144h) before sacrificing the mice. Control mice were reconstituted with BM of *Xcr1*-cre non-transgenic littermates and were injected with DT 12h before sacrificing the mice.

Tissue Sampling and Processing

When desired, red pulp DCs were labeled by injecting mice i.v. with 3 µg anti-CD11c antibody conjugated to APC (N418; BioLegend; 117310). Three minutes post-injection the mice were sacrificed by cervical dislocation or CO₂. For flow cytometry analysis, spleens and LNs were minced and digested in RPMI 1640 (Thermo Fisher Scientific; 21875-059) supplemented with recombinant DNase I (10 U/ml; Roche; 04 536 282 001) and Liberase TM (0.02 mg/ml; Roche; 05 401 127 001) at 37°C for 30min (spleen) or 20min (LN).

To dissect the thymus from mice, the mice were killed by CO₂. Thymocytes and splenocytes, as source for live cells at day8 in the cytotoxicity assay, were obtained by smashing the tissue on a 70 µm filter (Falcon).

In all cases, red blood cells were removed by osmotic lysis (homemade). The live cells were counted prior to antibody staining, by staining a sample with DAPI (Thermo Fisher Scientific;

D3571) or acridine orange/propidium iodide (Logos Biosystems; LB F23001) and counting with the BD FACSVerse (BD Bioscience) or LUNA-FX7 (Logos Biosystems), respectively.

Flow Cytometry and Cell Sorting

4 million live cells were stained with fluorescently and biotin labeled antibodies for staining of surface and intracellular markers. A first staining mixture consisted of Fc block (Polpharma Biologics) to avoid nonspecific binding, CD64-BV711 (X54-5/7.1; BioLegend; 139311) and CCR7-biotin (4B12; BioLegend; 13-1971-85) and was incubated at 4°C for 45min. A second staining step contained all other antibodies against surface markers and was performed at 4°C for 30min. In case intracellular markers needed to be stained, the cells were fixed and permeabilized using the Foxp3 Transcription Factor Staining Buffer Set (Thermo Fisher Scientific; 00-5523-00). Intracellular markers were stained at 4°C for 45 min. Viable cells were discriminated by Fixable Viability Dye eFluor 506 (Thermo Fisher Scientific; 65-0866-18) or eFluor 780 (Thermo Fisher Scientific; 65-0865-14). Biotinylated antibodies were conjugated to PE-CF594 or AF647 Streptavidin (BD Bioscience; 562284 and Thermo Fisher Scientific; S32357, respectively). In some experiments, cells were stained with BODIPY 493/503 (500 ng/ml; Thermo Fisher Scientific; D3922). The BODIPY 493/503 MFI was corrected by subtracting background fluorescence and normalized on the MFI of B cells, to minimize technical variance caused by the separate staining of different samples. Acquisition and analysis of labeled cell suspensions was performed with BD LSR Fortessa FACSsymphony A5 and A3 (BD Biosciences) cytometer equipped with FACSDiva software (BD Bioscience; v8.0.2). Single stained UltraComp eBeads (Thermo Fisher Scientific; 01-2222-42) and cells were prepared to adjust photomultiplier tube voltages to make sure the signal was within detection limits, to reduce fluorescence spill-over and to calculate the compensation matrix. A list of antibodies is provided in **Table S2**.

Cell sorting was performed on FACS ARIAI, ARIAI and FACSsymphony S6 sorter (BD Biosciences). To sort different cDC1 maturation subsets, the total single-cell suspension was enriched prior to staining and cell sorting by depletion of CD3e⁺, CD19⁺, Ly6G⁺, NK1.1⁺, TER119⁺ and CD64⁺ cells using biotin-labeled monoclonal antibodies and MagniSort Streptavidin Negative Selection Beads (Thermo Fisher Scientific; MSNB-6002-74).

Flow cytometry data was preprocessed by the PeacoQC algorithm (100)(code available at <https://github.com/saeyslab/PeacoQC>). This tool splits every fcs file in overlapping bins of equal amounts of cells and identifies whether there are any aberrant density peaks (based on an isolation tree and median absolute deviation). Measurement regions with such outlier values were removed from the fcs files before further downstream analysis. Flow data was analyzed with FlowJo10 software (FlowJo, BD).

RNA Isolation and RT-qPCR

Different cDC1 maturation subsets (5,000 – 50,000 cells) were sorted directly into RLT Plus buffer (Qiagen; 74034) supplemented with β -mercaptoethanol (1/100) and stored at –80 °C until further processing. RNA was obtained using the RNeasy Plus Micro Kit (Qiagen; 74034) according to manufacturer's protocol.

For RT-qPCR, complementary DNA was amplified using the Ovation PicoSL WTA System V2 kit (NuGEN; 2210-24), according to manufacturer's protocol. Pollutants were removed using the MinElute PCR purification kit (Qiagen; 28204), according to manufacturer's protocol, prior to SYBR Green-based (Bioline; BIO-98020) RT-qPCR (Lightcycler 480, Roche). mRNA expression was analyzed using qbase+ software version 3.2 (Biogazelle). GeNorm was used to select stable housekeeping genes. A list of primer sequences is provided in **Table S3**.

Cholesterol Quantification Assay

Distinct DC maturation subsets were sorted in FCS-containing buffer. After sorting, the DCs were washed with PBS and frozen as dry cell pellet at -150°C. Cholesterol was quantified using the Cholesterol/Cholesteryl Ester Quantitation Kit (Abnova; KA0829) according to manufacturer's instructions, except that the standard was diluted to range from 12.5 ng/well to 200 ng/well. The extract from at least 20,000 cells was used for the analysis. Total cholesterol (free and esterified) was measured. Fluorescence was measured with the FLUOstar omega (BMG Labtech).

Processing bulk RNA-seq data

RNA integrity was assessed using a Bioanalyser 2100 (Agilent). Sequencing was performed by VIB Nucleomics Core (www.nucleomics.be) on Illumina NextGenSeq platforms.

Preprocessing of the RNA-seq data was performed by Trimmomatic v0.35 (101) and quality control by FastQC v0.11.4. Mapping to the reference mouse genome was performed by Tophat2 v2.1.0 (102) and HTSeqCount v0.6.1 (103) was used for counting. Limma v3.42.2 (104) was used to normalize the data. Genes which did not meet the requirement of a count per million (cpm) value larger than 1 in at least 4 samples were filtered. This resulted in an expression table containing 11843 genes.

EdgeR v3.28.0 (105) was utilized to perform differential expression analysis. Benjamini-Hochberg correction was used to adjust the p-values for multiple testing. To be labelled as a DE gene, a gene needed to have an adjusted p-value smaller than 0.05 and a log2-ratio > 1 or < - 1.

The R package pheatmap v1.0.12 was used to create a heatmap of the top 100 DE genes between WT immature and mature cDC1s. The displayed gene expression was log2 normalized. The mean expression value per gene over all cell types was calculated and then subtracted from each cell type's particular gene expression value to scale the expression values.

Functional annotation of all the DE genes between WT immature and mature cDC1s was performed in Ingenuity Pathway Analysis software program (Qiagen). Z-scores, which indicate directionality, were linked to the log2-ratio of the genes. IPA core analysis was performed which leverages the Ingenuity Pathway Knowledge Base (IPKB) to identify affected canonical pathways, diseases and functions, and gene networks.

CITE-sequencing

Sorting and library prep

DCs were enriched as described earlier and 4 million cells were stained with FACS antibodies, TruStain FcX Block (BioLegend, 101320) and the mouse cell surface protein antibody panel containing 160 oligo-conjugated antibodies (TotalSeq-A, BioLegend) and 9 TotalSeq-A isotype controls (see Supplementary Table 2). 50.000 to 60.000 cDC1s (CD64⁺CD11c⁺MHCII⁺XCR1⁺CD172a⁻), 10.000 cDC2s (CD64⁺CD11c⁺MHCII⁺XCR1⁻CD172a⁺) and 10,000 random live cells were sorted.

Sorted single-cell suspensions were resuspended at an estimated final concentration of 1000 cells/μl and loaded on a Chromium GemCode Single Cell Instrument (10x Genomics) to generate single-cell gel beads-in-emulsion (GEM). The scRNA-Seq libraries were prepared using the GemCode Single Cell 3' Gel Bead and Library kit, version NextGEM 3.1 (10x Genomics) according to the manufacturer's instructions with the addition of amplification primer (3 nM, 5'CCTTGGCACCCGAGAATT*C*C) during cDNA amplification to enrich the TotalSeq-A cell surface protein oligos. Size selection with SPRIselect Reagent Kit

(Beckman Coulter) was used to separate amplified cDNA molecules for 3' gene expression and cell surface protein construction. TotalSeq-A protein library construction including sample index PCR using Illumina's Truseq Small RNA primer sets and SPRIselect size selection was performed according to the manufacturer's instructions. The cDNA content of pre-fragmentation and post-sample index PCR samples was analyzed using the 2100 BioAnalyzer (Agilent).

CITE-Sequencing Analysis

Sequencing libraries were loaded on an Illumina NovaSeq flow cell at VIB Nucleomics core with sequencing settings according to the recommendations of 10x Genomics, pooled in an 85:15 ratio for the combined 3' gene expression and cell surface protein samples, respectively.

The Cell Ranger pipeline (10x Genomics, v3.1.0) was used to perform sample demultiplexing and to generate FASTQ files for read 1, read 2 and the i7 sample index for the gene expression and cell surface protein libraries. Read 2 of the gene expression libraries was mapped to the reference genome (mouse mm10) using STAR. Subsequent barcode processing, unique molecular identifiers filtering and gene counting was performed using the Cell Ranger suite. CITE-seq reads were quantified using the feature-barcoding functionality. The average of the mean RNA reads per cell across both gene expression libraries was 17,435 reads, with an average sequencing saturation of 32.7%, as calculated by Cell Ranger. In total 2 individual WT single-cell libraries were created in this experiment, entailing 31,790 cells. The cellranger aggr pipeline was used to aggregate the 2 WT libraries into one feature-barcode matrix.

Pre-processing data

Pre-processing of the RNA UMI matrix was done by the scater (106) and scran R packages (v1.14.6) (106) according to the workflow proposed by the Marioni lab (107).

Outlier cells were identified based on three metrics (library size, number of expressed genes and mitochondrial proportion) and an initial lenient filtering was performed on these outlier cells according to median absolute deviation (MADs). Log-transformed normalized expression values were then computed from the count matrix.

Subsequently the Seurat R package (v3.1.4) (108) was used to create a Seurat object with both the raw counts and log2 transformed counts. SCTransform was performed on the raw count data to normalize the UMI counts by regularized negative binomial regression. Additionally, highly variable features were found and scaling was performed on the log-transformed count data as a back-up for SCT. Principal component analysis, clustering and tSNE/UMAP dimensionality reduction was performed. DoubletFinder (v2.0.2) (109) was performed to predict any remaining doublets in the dataset.

The ADT UMI matrix did not have extensive pre-processing performed. The same cells were filtered as during the RNA pre-processing to keep the columns of the matrices equal. The ADT expression data was processed using the Seurat pipeline, with CLR normalization and scaling of the data performed using the default parameters. Principal component analysis, clustering and tSNE/UMAP dimensionality reduction was performed too.

Antibodies with low expression in the studied samples were filtered out based on inspection of the density plots and feature plots for each antibody.

Marker genes per identified subpopulation were found using the findMarkers function of the Seurat pipeline and this informed the annotation process of the clusters.

After annotating the full dataset, a cDC1-subset and cDC2-subset were created and the same Seurat pipeline was performed as above. The DoubletFinder annotation together with ADT

data and information on mitochondrial proportion, UMI count and gene count was used to filter doublets, contaminating cells and low-quality clusters.

Nebulosa visualization

Some of the important features for cDC1 maturation seemed to be affected by the sparsity of single cell RNA-seq data, otherwise known as drop-out. To retrieve expression from these dropped-out features, they were visualized with Nebulosa (110), a visualization package in R (v1.1.0) based on kernel density estimation. It does this by incorporating the similarity between cells allowing a “convolution” of the cell features. The smoothing of the data significantly improves the visualization of the global expression of these features.

Trajectory inference

Dynverse was used to select the most appropriate method for trajectory inference.

Preparing the data, selecting the best method, inferring/visualizing/adapting the trajectory and performing differential expression on the trajectory was all performed with the dyno package (v0.1.2)(111).

The SCT data had multiple trajectories present (maturation and proliferation), so we opted for a method which could handle a multifurcating trajectory, PAGA Tree. This was a fairly complicated trajectory due to the nature of the dataset. To increase the interpretability, we adapted the trajectory and simplified it to the main maturation trajectory to determine the top DE genes linked to maturation.

The ADT data does not feature this proliferation loop, so we opted for a method suited for a linear trajectory, Scorpious.

Both SCT and ADT trajectories were projected onto the UMAP for easy interpretability.

CellCycleScoring Seurat

The cDC1 proliferation loop was split up into phases by calculating cell cycle phase scores based on canonical markers, from (112). The list consists of markers of the G2/M phase and the S phase and is loaded with Seurat. The human gene names were converted to mouse gene names by using the biomaRt package (v2.42.1) (113).

The scores from the CellCycleScoring function were stored in the object meta data. Based on these scores, cells could be assigned to either G2M, S or G1 phase.

Gene regulatory network analysis

The R package DoRothEA v1.2.2 (114) was used to perform gene regulatory network analysis (114). The tool provides a collection of mouse regulons which consist of transcription factors (TFs) with their transcriptional targets. It combines various data resources to attain this database of signed TF - target gene interactions and assigns varying confidence levels depending on the weight of the evidence found. This resource can be combined with a statistical method, VIPER (115), to infer TF activity from the mRNA expression of its transcriptional targets.

The mouse regulon data was pre-filtered to only contain the high confidence interactions (levels A/B/C). VIPER was then run on our cDC1 RNA expression data and the resulting inferred TF activity data was then stored inside the Seurat object as a separate data assay.

Statistical analysis

Statistical analyses and data representations were performed using Prism (GraphPad, La Jolla, CA). Data are presented as mean $\pm$ standard deviation (SD) or standard error of the mean

(SEM) as specified in the legend of the figures. Biological replicates are shown. The data distribution and variance characteristics were taken into account for statistical testing. In case of comparing more than two populations, ordinary one-way ANOVA with correction for multiple testing was used for the statistical analysis. Statistical significance was defined as $p < 0.05$ and two-sided testing was used. In some graphs, the statistical significance is not shown. Asterisks in graph represent the p-value unless mentioned otherwise. Number of biological replicates is indicated as dots in the figure or as 'n' in the legend. The investigator was not blinded to the mouse group allocation. The sample size was dependent on the availability or was determined by power analysis.

CITE-seq differential gene expression analysis was performed using the Wilcoxon Rank Sum test through the Seurat functions FindAllMarkers and FindMarkers. P-value adjustment was accomplished with Bonferroni correction. SCT and ADT markers for the annotated clusters were determined with default parameters, but only positive markers were evaluated. SCT and ADT transition markers were determined with a logfc.threshold of 0.20 and the results were subsequently filtered by a p_val_adj cut-off of 0.01. An extra "score" column was calculated to rank the importance of the genes as markers. It was calculated with this function: `"pct.1/(pct.2+0.01)*avg_logFC"`. The markers are ordered according to this score.

Supplementary Figures

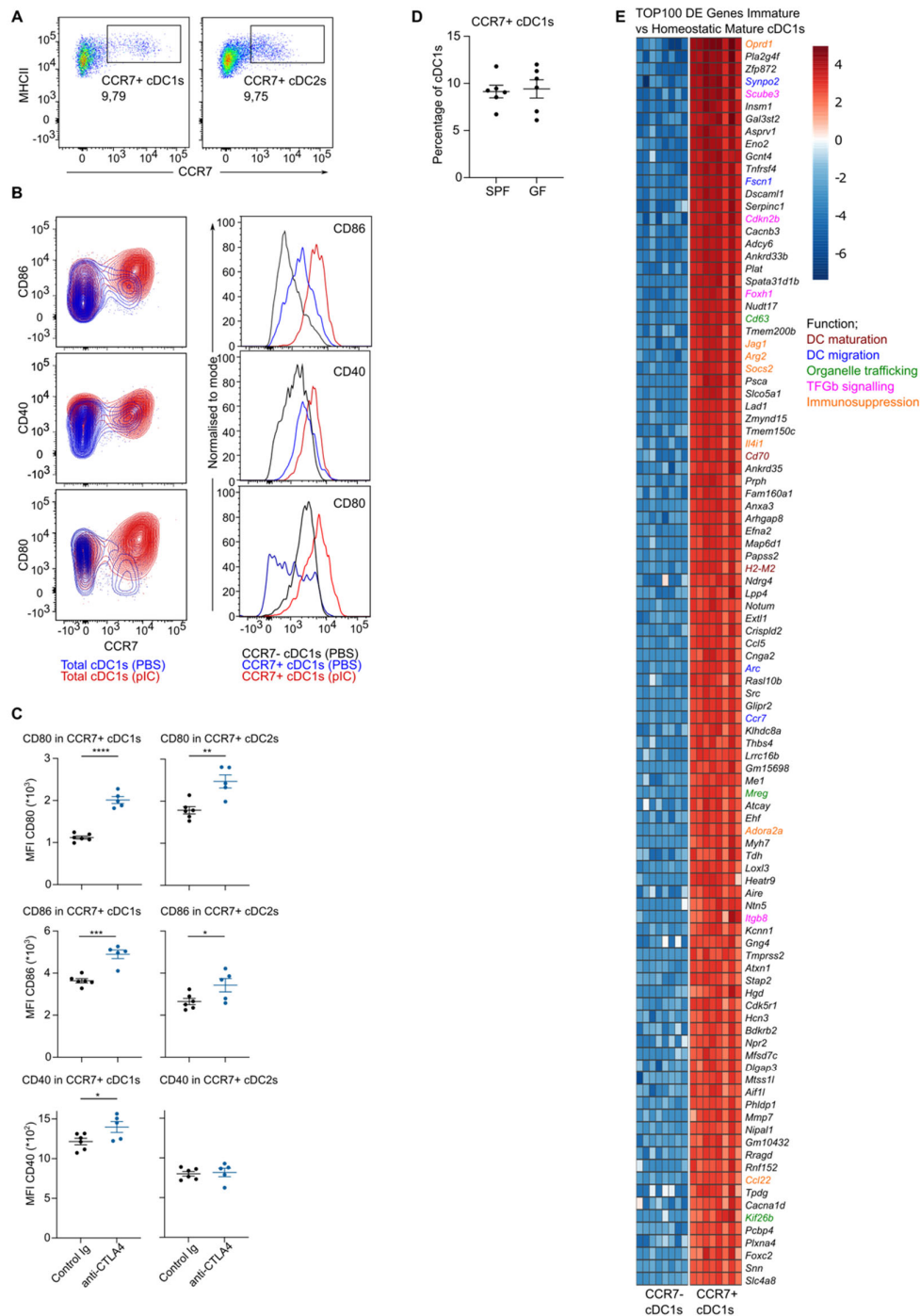


Figure S1. Homeostatic mature cDC1s express canonical DC maturation markers. (A) Representative staining profiles of CCR7⁺ cDC1s and cDC2s. (B) Representative surface expression of CD86, CD40 and CD80 on homeostatic immature CCR7⁻ cDC1s (black) and mature CCR7⁺ cDC1s (blue), and immunogenic mature CCR7⁺ cDC1s (red) after 16h i.v. poly(I:C) (pIC) (20 μ g) stimulation. (C) MFI of CD40, CD80 and CD86 in CCR7⁺ cDC1s and cDC2s after daily i.p. injection of 100 μ g anti-CTLA4 blocking antibody or isotype control for

3 days. Expression levels were determined on day 4. Unpaired t test. **(D)** The percentage of CCR7⁺ cDC1s in specific-pathogen free (SPF) and germ-free (GF) mice in homeostatic conditions. Unpaired t-test. **(E)** Heatmap of top 100 upregulated genes upon homeostatic maturation. Selected genes with well-established function in DC biology are highlighted and annotated to specific DC functions (colored). The color scale of the heatmap represents the scaled log2 normalized gene expression. Representative of at least 2 experiments in A-D. The mean $\pm$ SEM is shown in C and D. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$.

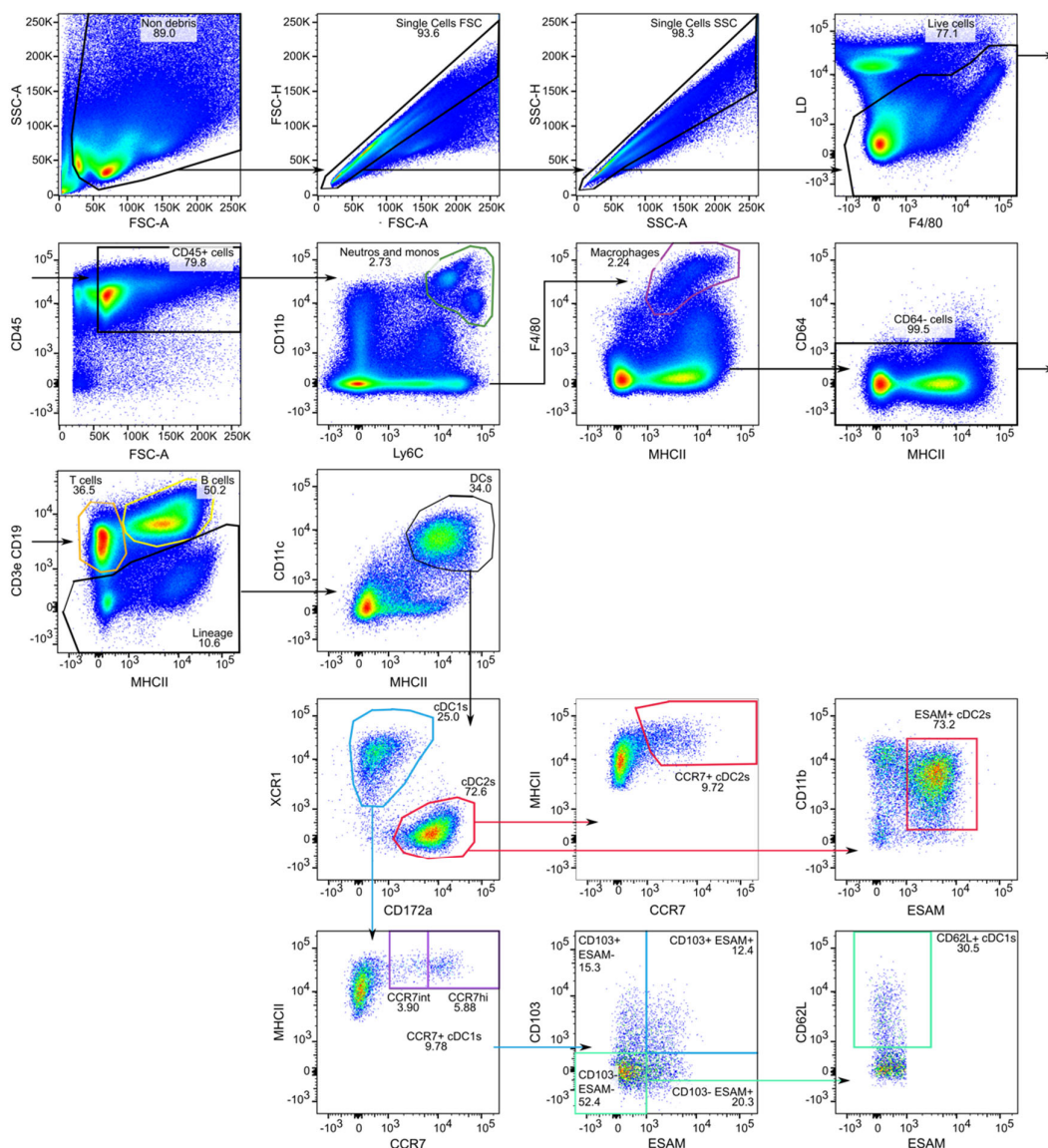


Figure S2. Representative gating strategy for the spleen. Gating on cDC1s subsets is indicated in blue. Gating on cDC2s is marked in red.

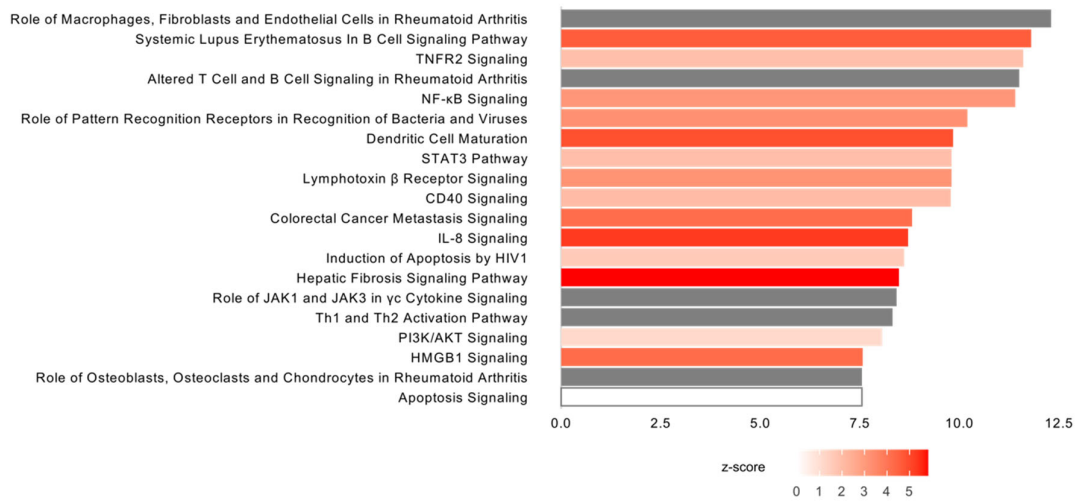


Figure S3. Top20 canonical pathways enriched in homeostatic matured cDC1s as determined by IPA. The x-axis shows the $-\log(p\text{-value})$. The color scale represents the z-score.

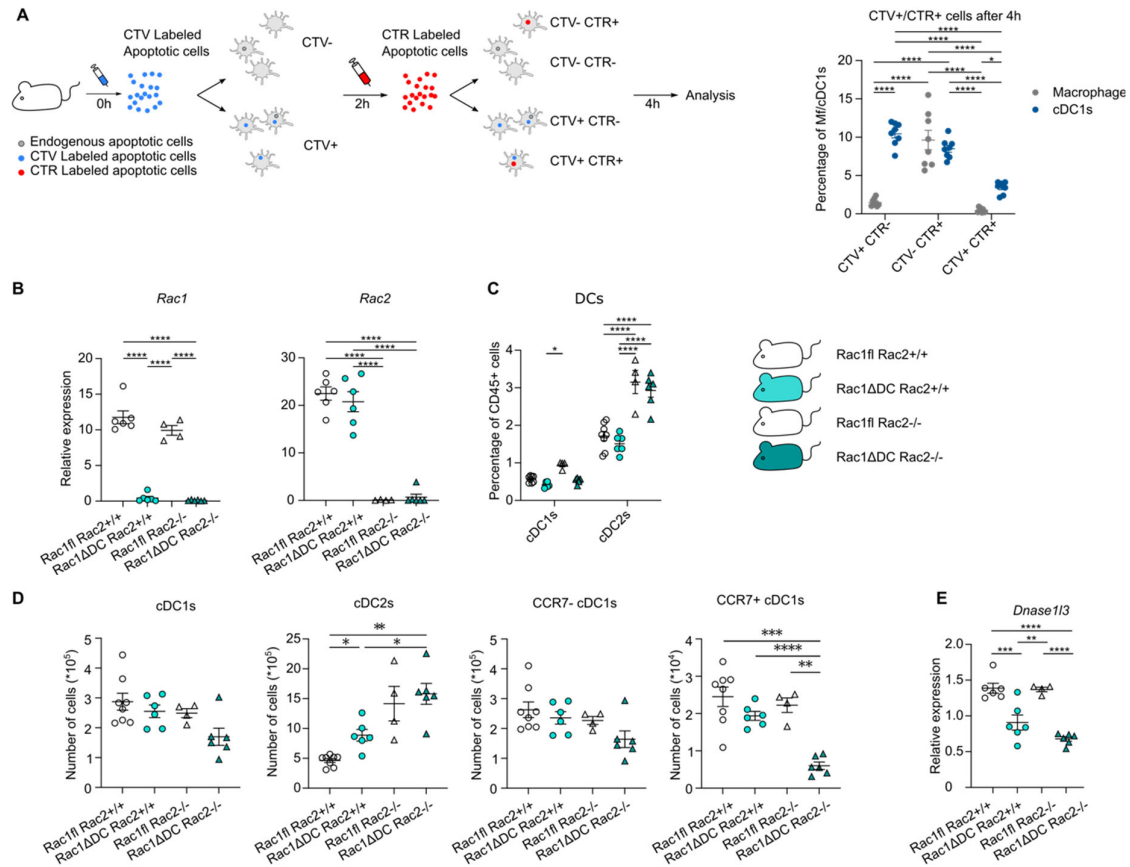


Figure S4. Secondary engulfment in macrophages and DCs. (A) To check whether CTV-labeled ACs are still available to be engulfed 4h after injection, mice were injected 2h after injection of CTV-labeled ACs with CTR-labeled ACs. The mice were sacrificed 2h after the second injection. The graph shows the percentage of macrophages and cDC1s that are CTV⁺ and/or CTR⁺ 4h after the first injection (i.e. 2h after the second injection). At that time point, macrophages were able to engulf CTR⁺ cells, while the CTV signal had completely disappeared. On the contrary, in cDC1s we detected both CTV⁺ cells (primary engulfer) as well as CTV⁺CTR⁺ double positive cells (secondary engulfer). Two Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. (B-C) Validation of *Rac1^{fl/fl}* or *ΔDC Rac2^{+/+}* or *Rac2^{-/-}* mice. (B) Expression of *Rac1*, *Rac2* in cDC1s is shown relative to housekeeping genes *Gapdh*, *Rpl13a* and *Ywhaz*. One Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. (C) Percentage of cDC1s and cDC2s in *Rac1^{fl/fl}* or *ΔDC Rac2^{+/+}* or *Rac2^{-/-}* mice. Two Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. (D) The number of total cDC1s and cDC2s, and CCR7⁻ and CCR7⁺ cDC1s in *Rac1^{fl/fl}* or *ΔDC Rac2^{+/+}* or *Rac2^{-/-}* mice. Welch ANOVA corrected for multiple testing by Dunnett's T3 multiple comparison. (E) Expression of *Dnase113* in cDC1s is shown relative to housekeeping genes *Gapdh*, *Rpl13a* and *Ywhaz*. One Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. The mean ± SEM is shown in A-D. Representative of at least 2 experiments in B-E. *p<0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001.

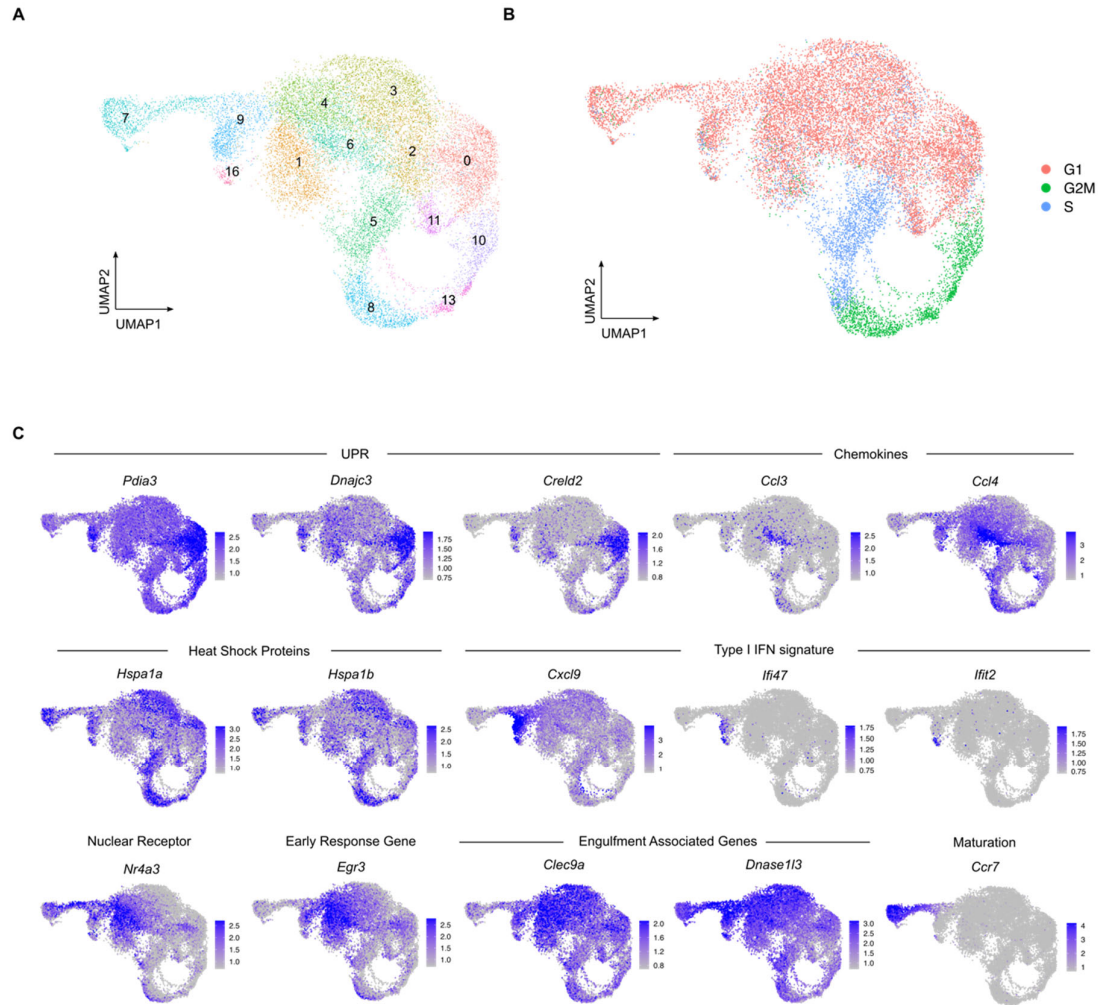


Figure S5. Details of the CITE-Seq analysis in cDC1s. (A) Annotation of CITE-seq clusters within cDC1s. Unsupervised clustering of the CITE-Seq data yielded 16 different clusters. Two were filtered out based on low quality UMI counts. Cluster 11 represents precDC1-like cells and was kept as such in the final annotation. Clusters 5, 8, 13 and 10 represent proliferating cDC1s in different phases of the cell cycle (see panel B) and were merged into ‘**proliferating cDC1s**’ in Fig. 3A. Clusters 0 and 2 are both CD62L⁺CD103⁻CD207⁻ESAM⁻ and were therefore merged into **early immature cDC1s**. Cluster 0 is hallmarked by a expression of UPR target genes (panel C: *Pdia3*, *Dnajc3*, *Creld2*), which are not detected in cluster 2. Clusters 3, 4, 6 and 1 are all CD62L⁻CD103⁺ESAM⁺CD207⁺CD63⁻CCR7⁻ and were merged as **late immature cDC1s**. Cluster 3 is positive for cytosolic heat shock proteins *Hspa1a* and *Hspa1b*. Especially cluster 4 and 6 are highly positive for the expression of Nr4a receptors, as well as *Egr* genes, which are expressed as a continuum from cluster 3 (Nr4a^{lo}) towards cluster 4 (Nr4a^{hi}). In this regard, cluster 3 could be considered as more immature than cluster 4, which is also suggested by the trajectories. Cluster 6 is clustered separately because of a high expression of *Ccl3* and *Ccl4*. Cluster 1 is clustered separately because of lower UMI values compared to the remaining clusters, which might have affected the expression of certain markers, but no separate genes are expressed, and most markers are very similar to the other late immature cDC1 subsets. Cluster 9 and 16 are both hallmarked by expression of IFN responsive genes. Some of them, especially *Ifi* genes, are unique to cluster 16. Others (like

Cxcl9 and *Cxcl10*) appear expressed in both clusters. Surface markers of cluster 16 and cluster 9 are distinct (cluster 16 more positive for CD62L and CD43, negative for CD207). We decided to merge them for simplicity, although it remains possible that cluster 16 represents a distinct, more immature subset that also shows a type I IFN signature. As can be seen in the feature plots, several features are differentially expressed between the lower dense cloud of cluster 9 compared to the top part (the part that forms a continuum with late mature cells). In an independent experiment, these two parts were further split by the clustering algorithm. **(B)** UMAP of cell cycle stages. The color codes highlight different cell cycle stages. **(C)** Feature plots showing the expression of representative genes of the UPR, maturation, Heat shock proteins, Nuclear Receptors, chemokines, Type I IFN inducible genes and engulfment associated genes. The color scale represents normalized gene expression. Representative of 2 experiments.

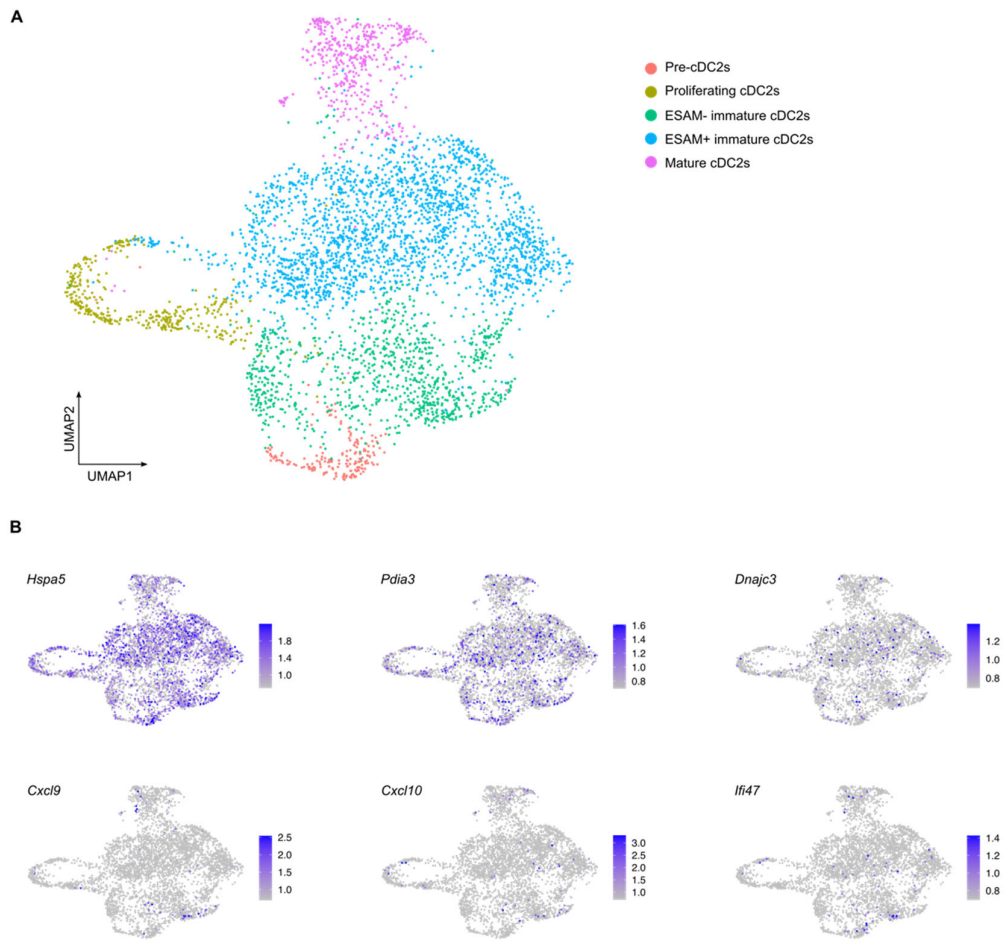


Figure S6. CITE-Seq analysis in cDC2s. (A) UMAP plots of CITE-seq data of cDC2s. The colors represent the annotated cDC2 clusters. (B) UMAP feature plots showing the expression of genes involved in the UPR and genes inducible by type I IFN. The color scale represents normalized gene expression. Representative of 2 experiments.

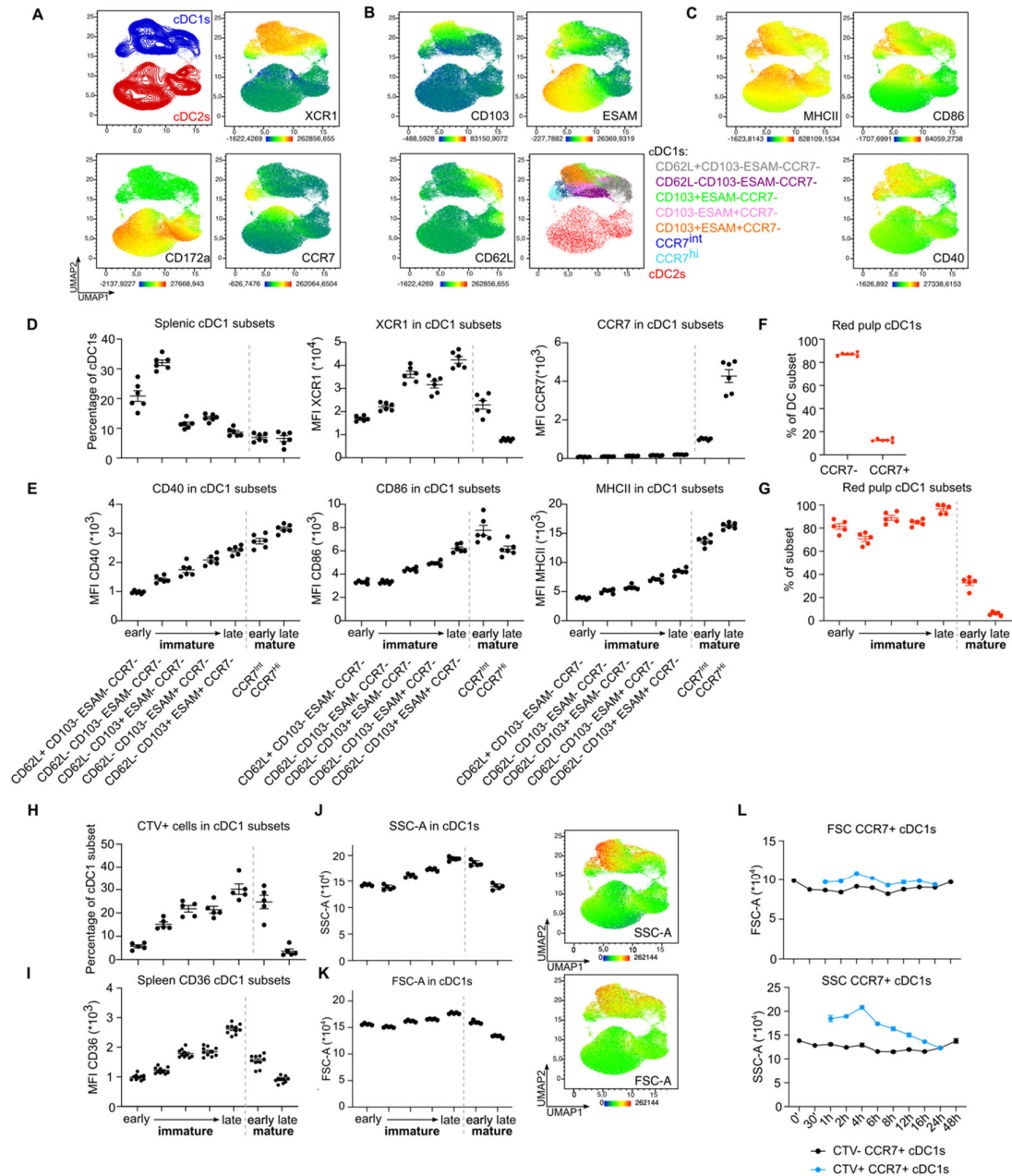


Figure S7. Analysis of distinct cDC1 maturation stages by flow cytometry. (A-C) UMAP plots of flow cytometry data of pooled lineage⁻ CD11c⁺ MHCII⁺ cells showing the density (contour plot), expression of indicated surface markers and overlay with manually gated cDC1 subsets. The color scale represents the MFI. (D-E) Distribution of cDC1 subsets and expression of indicated surface markers on manually gated cDC1 subsets. The markers chosen for manual gating were chosen based on the CITE-seq data. The x-axes contain the schematic representation used in the main figures and the detailed marker expression of each subset. (F-G) Frequency of total cDC1s, cDC2s and cDC1 subsets in the red pulp/marginal zone. Red pulp/marginal zone DCs were stained by injecting mice i.v. with 3 μ g CD11c-APC antibody and sacrificing them after 3 minutes. (H) Uptake of CTV-labeled ACs in cDC1 subsets, measured by CTV signal 2h after injection. (I) Expression of CD36 as MFI in cDC1 subsets, measured by flow cytometry. (J-K) Forward and side scatter area in cDC1 subsets. UMAP

plots of flow cytometry data of pooled lineage CD11c⁺ MHCII⁺ cells showing the forward and side scatter area in DCs. (L) Kinetics of FSC-A and SSC-A on CTV⁻ and CTV⁺ CCR7⁺ cDC1s in mice after i.v. injection with CTV-labeled ACs. N = 1 for the 0' time point, n = 5 for all other time points. The X-axis in G-K shows the schematic representation of cDC1 subsets. The mean $\pm$ SEM is shown in D-L. Representative of at least 2 experiments in A-L.

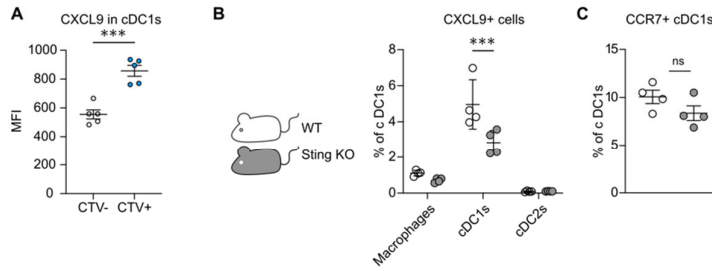


Figure S8. The STING pathway is not needed for homeostatic cDC1 maturation. (A) Expression of CXCL9 in CTV⁻ and CTV⁺ cDC1s 2h after injection of CTV-labeled apoptotic thymocytes, measured by flow cytometry. Paired t-test. (B) The percentage of CXCL9⁺ cells in Macrophages, cDC1s and cDC2s in wild-type (WT) and STING knock-out (KO) mice by flow cytometry. Two-way ANOVA with multiple comparisons test controlling the False Discovery Rate. Asterisks represent the q-value. (C) The percentage of CCR7⁺ cDC1s in wild-type (WT) and STING knock-out (KO) mice by flow cytometry. Unpaired t-test. *p<0.05; **p<0.01; ***p<0.001; ****p<0.0001; ns, not significant. The mean ± SEM is shown.

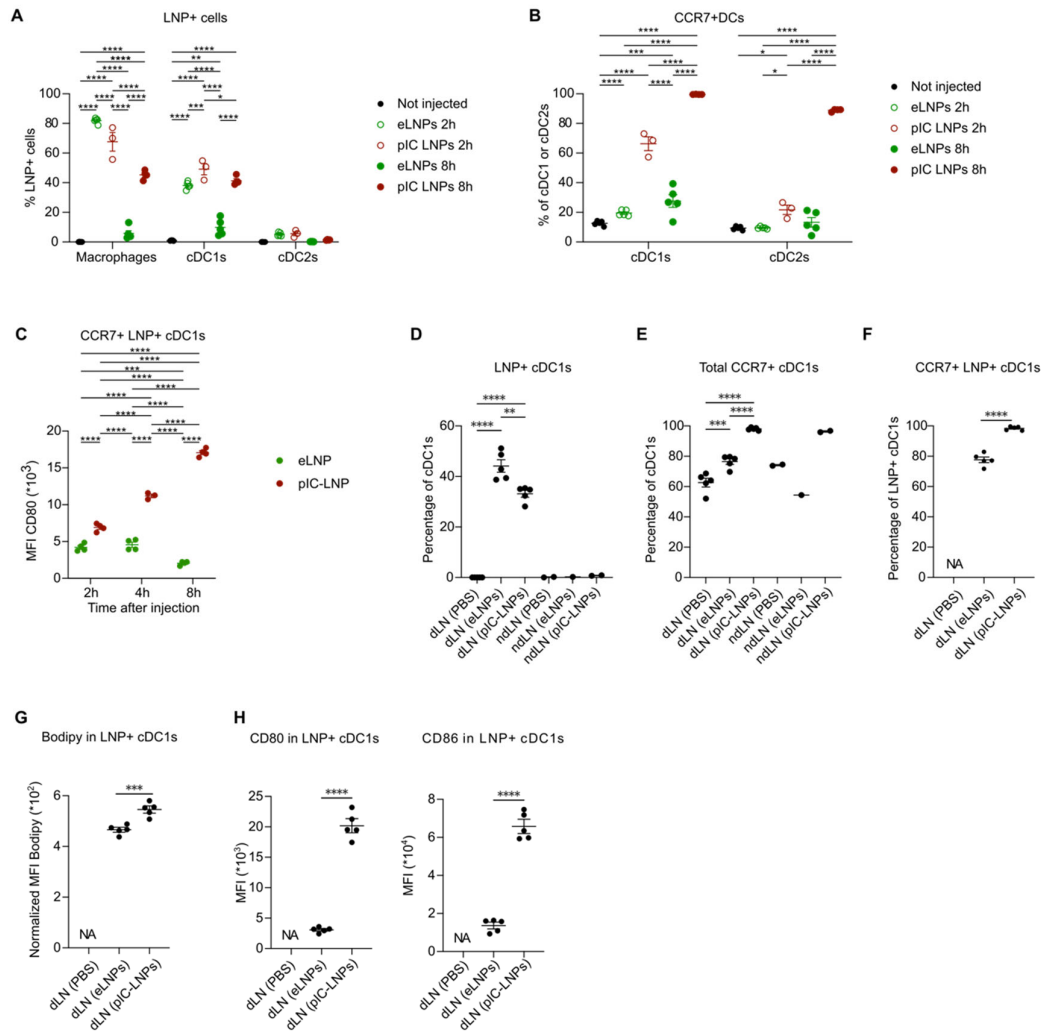


Figure S9. Analysis of LNP uptake in different immune cell types in the spleen and lymph node. (A) The percentage of LNP⁺ Macrophages, cDC1s and cDC2s 2h and 8h after Cy5-labeled eLNP and pIC-LNP i.v. injection. Two Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. (B) The percentage of CCR7⁺ cDC1s and cDC2s of total cDC1s and cDC2s, respectively, 2h and 8h after Cy5-labeled eLNP and pIC-LNP i.v. injection. Representative of at least 2 experiments in A-B. Two Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. (C) Expression analysis of CD80 by flow cytometry in CCR7⁺LNP⁺ cDC1s at different time points post-injection with eLNPs versus pIC-coupled LNPs. Two Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. (D) Percentage of LNP⁺ cDC1s in the draining (dLN, iliac LN) and non-draining LN (ndLN, inguinal LN) 24h after intramuscular injection of DiD-labeled eLNPs and pIC-LNPs. One Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. (E) Percentage of total mature cDC1s in the dLN and ndLN 24h after intramuscular injection of DiD-labeled eLNPs and pIC-LNPs. One Way Anova, corrected for multiple testing by Tukey's multiple comparison test with a single pooled variance. (F) Percentage of CCR7⁺ cDC1s within the LNP⁺ cDC1s in the dLN 24h after intramuscular injection of DiD-labeled eLNPs and pIC-LNPs. Unpaired t-test with pooled variance. (G)

BODIPY staining of LNP⁺ cDC1s in the dLN 24h after intramuscular injection of DiD-labeled eLNPs and pIC-LNPs. Unpaired t-test with pooled variance. **(H)** CD80 and CD86 levels in LNP⁺ cDC1s in the dLN measured by flow cytometry 24h after intramuscular injection of DiD-labeled eLNPs and pIC-LNPs. Unpaired t-test with pooled variance. Statistics for the ndLN are omitted. *p<0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001. The mean ± SEM is shown. Representative of at least 2 experiments in A-F.

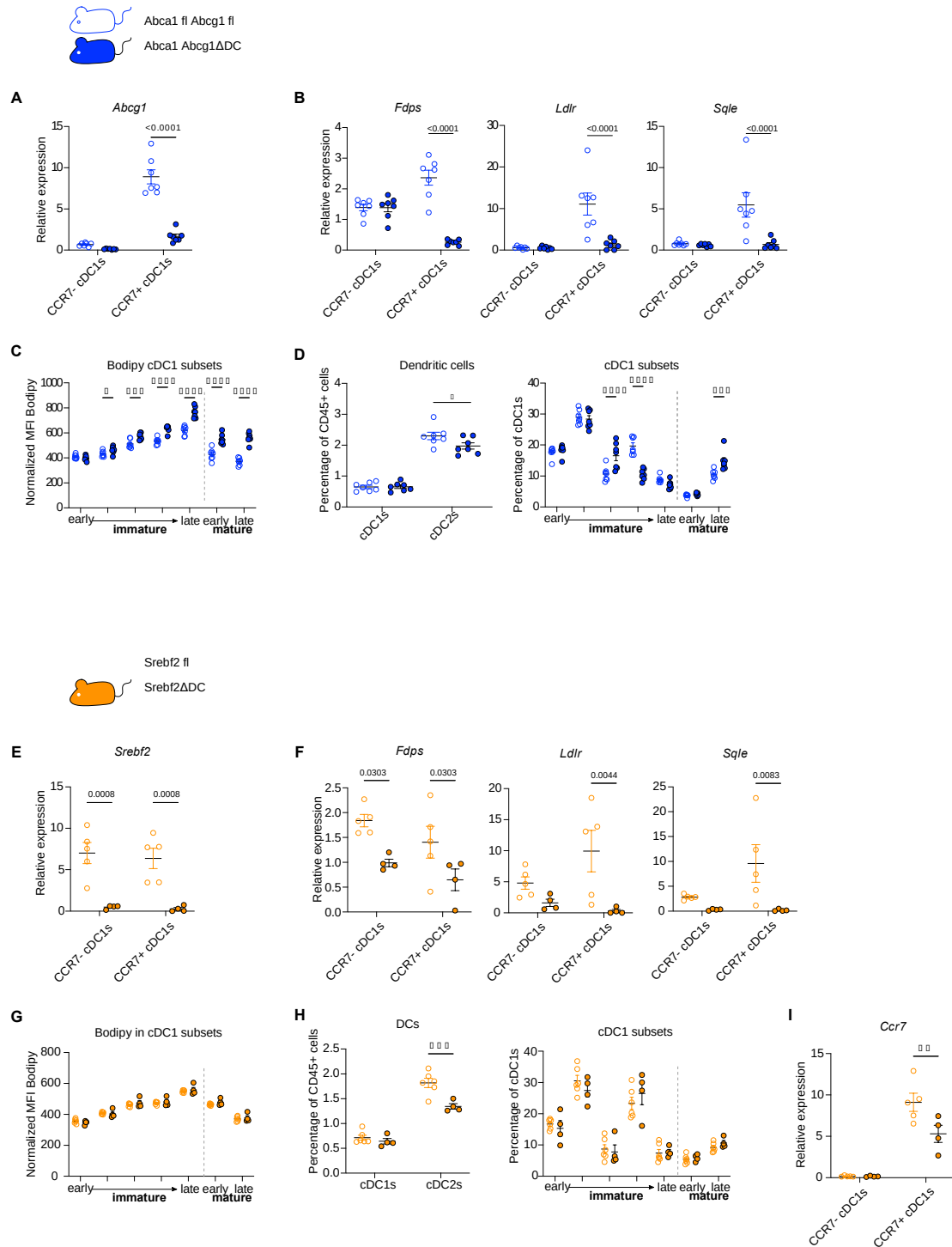


Figure S10. Neither the ABCA1/ABCG1 nor the SREBP2 pathway are needed for inducing the homeostatic cDC1 maturation program. (A) Validation of the *Abca1*^{fl/fl} *Abcg1*^{fl/fl} / *Abca1**Abcg1*ΔDC mice by qPCR. *Abca1* expression could not be detected by qPCR, probably due to its very low expression in DCs. The expression was normalized to housekeeping genes *Gapdh* and *Rpl13a*. Two-way ANOVA with multiple comparison test with single pooled variance, corrected by controlling the False Discovery Rate. (B) Activation of cholesterol metabolic genes in immature and mature cDC1s isolated from *Abca1*^{fl/fl} *Abcg1*^{fl/fl} / *Abca1**Abcg1*ΔDC mice measured by qPCR. The expression was normalized to housekeeping genes *Gapdh* and *Rpl13a*. Two-way ANOVA with multiple comparison test with single pooled variance, corrected by controlling the False Discovery Rate. (C) BODIPY staining in different

cDC1 subsets of *Abca1*^{fl/fl} *Abcg1*^{fl/fl} / *Abca1Abcg1*ΔDC mice. Two-way ANOVA with multiple comparison test with single pooled variance, corrected by controlling the False Discovery Rate. Asterisks represent the q-value. **(D)** Percentage of cDC1s and cDC2 subsets in *Abca1*^{fl/fl} *Abcg1*^{fl/fl} / *Abca1Abcg1*ΔDC mice. Two-way ANOVA with multiple comparison test with single pooled variance, corrected by controlling the False Discovery Rate. Asterisks represent the q-value. **(E)** Validation of the *Srebf2*^{fl/fl} / *Srebf2*ΔDC mice by qPCR. The expression was normalized to housekeeping genes *Gapdh*, *Rpl13a*, and *Ywhaz*. Two-way ANOVA with multiple comparison test with single pooled variance, corrected by controlling the False Discovery Rate. Asterisks represent the q-value. **(F)** Activation of cholesterol metabolic genes in immature and mature cDC1s isolated from *Srebf2*^{fl/fl} / *Srebf2*ΔDC mice measured by qPCR. The expression was normalized to housekeeping genes *Gapdh*, *Rpl13a*, and *Ywhaz*. Two-way ANOVA with multiple comparison test with single pooled variance, corrected by controlling the False Discovery Rate. **(G)** BODIPY staining in different cDC1 subsets of *Srebf2*^{fl/fl} / *Srebf2*ΔDC mice. Two Way Anova, corrected for multiple testing by controlling the False Discovery Rate. **(H)** Percentage of cDC1s, cDC2s and cDC1 subsets in *Srebf2*^{fl/fl} / *Srebf2*ΔDC mice. Two Way Anova, corrected for multiple testing by controlling the False Discovery Rate. **(I)** Expression of CCR7 in immature and mature cDC1s isolated from *Srebf2*^{fl/fl} / *Srebf2*ΔDC mice measured by qPCR. The expression was normalized to housekeeping genes *Gapdh*, *Rpl13a*, and *Ywhaz*. Two-way ANOVA with multiple comparison test with single pooled variance, corrected by controlling the False Discovery Rate. Asterisks represent the q-value. *p<0.05;**p < 0.01; ***p < 0.001; ****p < 0.0001. The mean ± SEM is shown. Representative of at least 2 experiments.

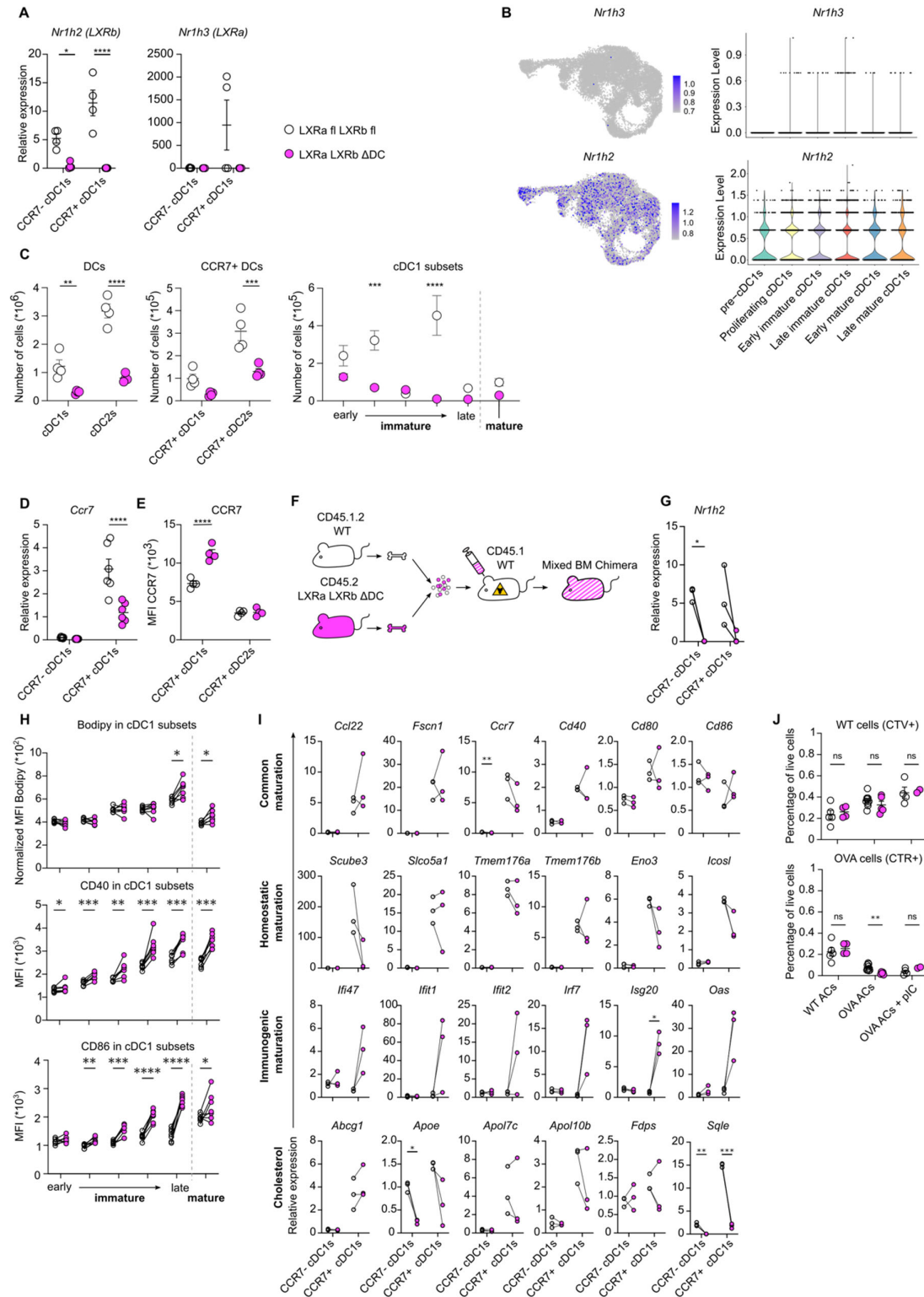


Figure S11. Analysis of the $LXR\alpha LXR\beta \Delta DC$ mice and $LXR\alpha/\beta$ chimeric mice. (A) Validation of $LXR\alpha^{\Delta/\Delta} LXR\beta^{\Delta/\Delta} / LXR\alpha LXR\beta \Delta DC$ mice. Expression of *Nr1h2* and *Nr1h3* in CCR7 $^{-}$ and CCR7 $^{+}$ cDC1s is shown relative to housekeeping genes *Sdha*, *Rpl13a* and *Ywhaz*. Two Way Anova, corrected for multiple testing by Šidák's multiple comparisons test with a

single pooled variance. **(B)** UMAP feature plots and violin plots of CITE-seq data showing the expression of Nr1h2 and Nr1h3 in cDC1s. **(C)** Number of cDC1s, cDC2s and cDC1 subsets of LXR $\alpha^{\text{fl/fl}}$ LXR $\beta^{\text{fl/fl}}$ or LXR α LXR $\beta\Delta$ DC mice. Two Way Anova, corrected for multiple testing by Šídák's multiple comparisons test with a single pooled variance (left and middle), with individual variances (right). **(D)** Expression of Ccr7 mRNA levels in CCR7 $^{-}$ and CCR7 $^{+}$ cDC1s from LXR $\alpha^{\text{fl/fl}}$ LXR $\beta^{\text{fl/fl}}$ or LXR α LXR $\beta\Delta$ DC mice is shown relative to housekeeping genes *Sdha*, *Rpl13a* and *Ywhaz*. Two Way Anova, corrected for multiple testing by Šídák's multiple comparisons test with a single pooled variance. **(E)** Flow cytometry analysis of CCR7 protein expression in CCR7 $^{+}$ cDC1s and cDC2s from LXR $\alpha^{\text{fl/fl}}$ LXR $\beta^{\text{fl/fl}}$ or LXR α LXR $\beta\Delta$ DC mice. Two Way Anova, corrected for multiple testing by Tukey's multiple comparison test with individual variances. **(F)** CD45.1 acceptor mice were irradiated with 8 Gy. 6h later the mice were reconstituted with bone marrow of CD45.1.2 wild-type mice and CD45.2 LXR α LXR $\beta\Delta$ DC mice in a 1:1 ratio. 2 months later the mice were sacrificed for analysis. **(G)** Relative expression of Nr1h2 in CCR7 $^{-}$ and CCR7 $^{+}$ cDC1s coming from either the wild-type or LXR α LXR $\beta\Delta$ DC part of the bone marrow, measured by RT-qPCR. The expression was normalized to housekeeping genes *Gapdh*, *Sdha* and *Ywhaz*. Multiple paired t-test, correcting for multiple testing by controlling the False Discovery Rate. Asterisks represent the q-value. **(H)** The normalized MFI of BODIPY 493/503 and MFIs of CD40 and CD86 on cDC1 subsets from LXR $\alpha^{\text{fl/fl}}$ LXR $\beta^{\text{fl/fl}}$ or LXR α LXR $\beta\Delta$ DC mice. Multiple paired t-test, correcting for multiple testing by controlling the False Discovery Rate. Asterisks represent the q-value. **(I)** Relative expression of common maturation markers (first row), homeostatic maturation markers (second row), immunogenic maturation markers (third row) and cholesterol efflux/biosynthesis genes (last row) in CCR7 $^{-}$ and CCR7 $^{+}$ cDC1s coming from either the wild-type or LXR α LXR $\beta\Delta$ DC part of the bone marrow, measured by RT-qPCR. The expression was normalized to housekeeping genes *Gapdh*, *Sdha* and *Ywhaz*. Multiple paired t-test, correcting for multiple testing by controlling the False Discovery Rate. Asterisks represent the q-value. Lines connect paired values. **(J)** Percentage of CTV-labeled WT (top) and CTR-labeled OVA (bottom) live target cells in wild-type or LXR α LXR $\beta\Delta$ DC mice. Multiple Mann-Whitney U tests, corrected for multiple testing by controlling the False Discovery Rate. Asterisks represent the q-value. *p<0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001. The mean $\pm$ SEM is shown. Representative of 2 experiments in A-E.

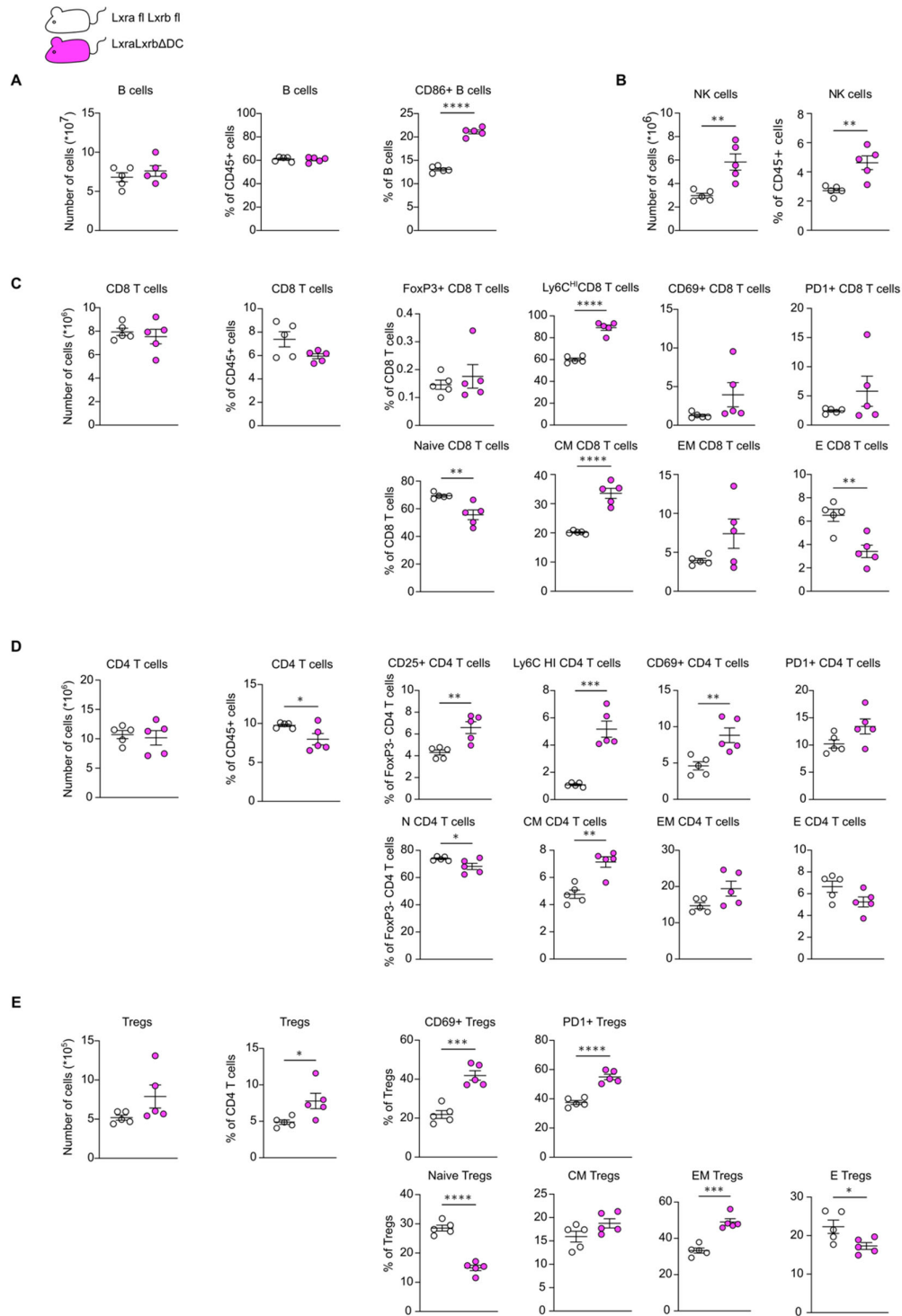


Figure S12. Immunophenotyping $LXR\alpha LXR\beta\Delta DC$ mice reveals an inflammatory phenotype. Flow cytometry analysis of $LXR\alpha^{fl/fl} LXR\beta^{fl/fl} / LXR\alpha LXR\beta\Delta DC$ mice. Unpaired t-test. The mean $\pm$ SEM is shown. Representative of two experiments. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$. The mean $\pm$ SEM is shown.

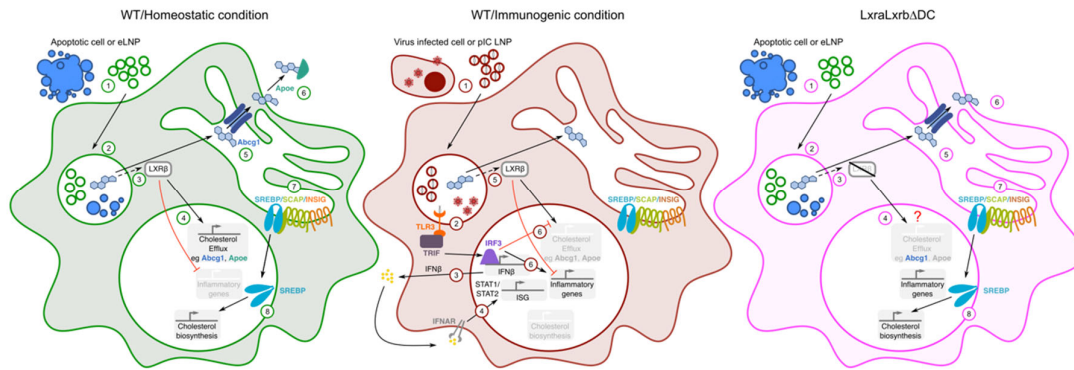


Figure S13: Model depicting the regulation of cholesterol metabolic pathways in different conditions in cDC1s. *Left panel:* Upon engulfment of apoptotic cells or empty LNPs (1) by cDC1s and concomitant influx of cholesterol (2), LXR β becomes activated (3) and triggers cholesterol efflux genes (4-6). At the same time, LXR β represses expression of inflammatory genes (4), possibly in an NCoR-dependent manner (85). While the SREBP2 pathway is repressed at the peak of cholesterol influx, the pathway becomes activated at the final step of tolerogenic DC maturation in an Abcg1-dependent manner. Based on literature, this might be caused by an Abcg1-dependent shuttling of sterols out of the endoplasmic reticulum, leading to a drop in ER sterol levels and reactivation of the SREBP2 pathway (75, 76) (7). SREBP2 causes activation of a cholesterol biosynthesis program (8). *Middle panel:* Upon engulfment of pIC-coupled LNPs or virus-infected cells (1), pIC will activate TLR3 on endosomes (2), which will lead to the activation of a TRIF/IRF3 pathway and release of type I IFN (3). Secreted IFN β will bind IFNAR receptors on the cell surface of DCs and activate a type I IFN response, leading to rigorous stimulation of IFN stimulated genes (ISGs)(4). Influx of cholesterol will also trigger LXR pathways (5). However, LXR signaling is strongly repressed likely in an IRF3-dependent manner (100). Hence, cholesterol efflux genes are not expressed neither are inflammatory genes repressed in an LXR-dependent manner (6). Cholesterol accumulates in the cell, Abcg1 is not induced and the SREBP2 pathway is not activated. *Right panel:* In LXR α LXR β Δ DC cDC1s, uptake of LNPs (1) leads to influx of cholesterol (2). Due to the genetic deficiency of LXRs (3), LXR-driven cholesterol efflux genes like *ApoE* are not induced (4). Additionally, basal activation of inflammatory genes is no longer repressed in an LXR-dependent manner (4). Still, the *Abcg1* gene appears induced by an as-yet unknown pathway (5). We postulate that the continued presence of Abcg1 allows for reshuttling of sterols out of the ER (6), hence enabling activation of the SREBP2 program (7). Therefore, cholesterol biosynthesis genes are still being turned on in mature LXR α LXR β Δ DC CCR7 $^{+}$ cDC1s (8).

Table S1. Composition of empty LNPs and pIC-coupled LNPs

	PIC-LNP	EMPTY LNP
N/P RATIO*	5:1	5:1
AMOUNT PIC (MG)	0.75	/
IONIZABLE LIPID (MOL%)	50	50
CHOLESTEROL (MOL%)	38.5	38.5
DOPE (MOL%)	10	10
DMG-PEG (MOL%)	1.5	1.5
DOPE-CY5 (MOL%) (SUBTRACTED FROM DOPE)	0.25	0.25
DID	0.25	0.25

*Target molar charge ratio (mol ionizable lipid: mol pIC)

Table S2. Overview of all antibodies used for flow cytometry and/or FACS

ANTIBODIES	SOURCE	IDENTIFIER
Armenian Hamster anti-mouse CD11c (clone N418), APC conjugated	BioLegend	Cat# 117310
Armenian Hamster anti-mouse CD11c (clone N418), eFluor450 conjugated	Thermo Fisher Scientific	Cat# 48-0114-82
Hamster anti-mouse CD11c (clone HL3), FITC conjugated	BD Bioscience	Cat# 553801
Hamster anti-mouse CD11c (clone HL3), BUV737 conjugated	BD Bioscience	Cat# 612796
Rat anti-mouse CD86 (clone GL1), PE conjugated	BD Bioscience	Cat# 553692
Rat anti-mouse CD86 (clone GL1), BV421 conjugated	BioLegend	Cat# 105032
Rat anti-mouse CD40 (clone 3/23), APC conjugated	BioLegend	Cat# 124612
Armenian Hamster anti-mouse CD80 (clone 16-10A1), PE conjugated	BD Bioscience	Cat# 553769
Armenian Hamster anti-mouse CD80 (clone 16-10A1), BUV737 conjugated	BD Bioscience	Cat# 612773
Rat anti-mouse CD197 (clone 4B12), biotin conjugated	Thermo Fisher Scientific	Cat# 13-1971-85
Rat anti-mouse I-A/I-E (clone M5/114.15.2), APC-eFluor780 conjugated	Thermo Fisher Scientific	Cat# 47-5321-82
Rat anti-mouse I-A/I-E (clone M5/114.15.2), FITC conjugated	Thermo Fisher Scientific	Cat# 11-5321-85
Rat anti-mouse I-A/I-E (clone M5/114.15.2), BUV805 conjugated	BD Bioscience	Cat# 748844
Rat anti-mouse I-A/I-E (clone M5/114.15.2), BUV496 conjugated	BD Bioscience	Cat# 750281
Rat anti-mouse CD11b (clone M1/70), BUV395 conjugated	BD Bioscience	Cat# 563553
Rat anti-mouse CD11b (clone M1/70), APC-R700 conjugated	BD Bioscience	Cat# 564985
Rat anti-mouse Ly-6G (clone 1A8), AF700 conjugated	BioLegend	Cat# 127616
Rat anti-mouse Ly-6G (clone 1A8), biotin conjugated	BioLegend	Cat# 127604
Rat anti-mouse Ly-6C (clone HK1.4), PerCP-Cy5.5 conjugated	Thermo Fisher Scientific	Cat# 45-5932-82
Rat anti-mouse Ly-6C (clone HK1.4), BV570 conjugated	BioLegend	Cat# 128030
Mouse anti-mouse CD64 (clone X54-5/7.1), BV711 conjugated	BioLegend	Cat# X54-5/7.1
Mouse anti-mouse CD64 (clone X54-5/7.1), BV421 conjugated	BioLegend	Cat# 139309
Mouse anti-mouse CD64 (clone X54-5/7.1), biotin conjugated	BioLegend	Cat# 139318
Armenian Hamster anti-mouse CD3e (clone 145-2c11), BUV395 conjugated	BD Bioscience	Cat# 563565
Armenian Hamster anti-mouse CD3e (clone 145-2c11), BV605 conjugated	BioLegend	Cat# 100351
Armenian Hamster anti-mouse CD3e (clone 145-2c11), PE-Cy5 conjugated	Thermo Fisher Scientific	Cat# 15-0031-83
Armenian Hamster anti-mouse CD3e (clone 145-2c11), APC-Cy7 conjugated	BD Bioscience	Cat# 557596
Armenian Hamster anti-mouse CD3e (clone 145-2c11), biotin conjugated	Thermo Fisher Scientific	Cat# 13-0031-85
Rat anti-mouse CD19 (clone 1D3), BUV563 conjugated	BD Bioscience	Cat# 749028
Rat anti-mouse CD19 (clone 6D5), BV605 conjugated	BioLegend	Cat# 115540
Rat anti-mouse CD19 (clone 1D3), PE-Cy5 conjugated	Thermo Fisher Scientific	Cat# 15-0193-83
Rat anti-mouse CD19 (clone 1D3), APC-Cy7 conjugated	BD Bioscience	Cat# 557655
Rat anti-mouse CD19 (clone 6D5), BV786 conjugated	BioLegend	Cat# 115543
Rat anti-mouse CD19 (clone 1D3), biotin conjugated	Thermo Fisher Scientific	Cat# 13-0193-85

Armenian Hamster anti-mouse TCR β (clone H57-597), BV786 conjugated	BioLegend	Cat# 109249
Rat anti-mouse CD135 (clone A2F10), PE conjugated	Thermo Fisher Scientific	Cat# 12-1351-83
Rat anti-mouse CD135 (clone A2F10), APC conjugated	BD Bioscience	Cat# 560718
Rat anti-mouse CD26 (clone H194-112), FITC conjugated	BD Bioscience	Cat# 559652
Mouse anti-mouse XCR1 (clone ZET), BV510 conjugated	BioLegend	Cat# 148218
Mouse anti-mouse XCR1 (clone ZET), BV650 conjugated	BioLegend	Cat# 148220
Mouse anti-mouse XCR1 (clone ZET), PE conjugated	BioLegend	Cat# 148204
Rat anti-mouse CD172a (clone P84), PerCP-eFluor710 conjugated	Thermo Fisher Scientific	Cat# 46-1721-82
Rat anti-mouse CD172a (clone P84), PE-Cy7 conjugated	BioLegend	Cat# 144008
Rat anti-mouse ESAM (clone 1G8), PE conjugated	Thermo Fisher Scientific	Cat# 12-5852-82
Rat anti-mouse ESAM (clone 1G8), PE-Cy7 conjugated	BioLegend	Cat# 136212
Rat anti-mouse CD103 (clone M290), BUV395 conjugated	BD Bioscience	Cat# 740238
Rat anti-mouse CD62L (clone MEL-14), PE-Cy5 conjugated	BioLegend	Cat# 104410
Rat anti-mouse CD62L (clone MEL-14), BV421 conjugated	BioLegend	Cat# 104436
Rat anti-mouse CD207 (clone 929F3.01), AF488 conjugated	Dendritics	Cat# DDX0362A488-100
Rat anti-mouse CD207 (clone 929F3.01), AF647 conjugated	Dendritics	Cat# DDX0362A647-100
Mouse anti-mouse CD36 (clone CRF D-2712), APC conjugated	BD Bioscience	Cat# 562744
Rat anti-mouse F4/80 (clone BM8), BV786 conjugated	BioLegend	Cat# 123141
Mouse anti-mouse CD45.1 (clone A20), FITC conjugated	Thermo Fisher Scientific	Cat# 11-0453-85
Mouse anti-mouse CD45.2 (clone 104), BUV737 conjugated	BD Bioscience	Cat# 612778
Rat anti-mouse CD45 (clone 30-F11), FITC conjugated	BioLegend	Cat# 103108
Rat anti-mouse CD45 (clone 30-F11), BV480 conjugated	BD Bioscience	Cat# 566095
Rat anti-mouse CD45 (clone 30-F11), BV605 conjugated	BD Bioscience	Cat# 563053
Rat anti-mouse CD45 (clone 30-F11), BUV496 conjugated	BD Bioscience	Cat# 749889
Rat anti-mouse CD45 (clone 30-F11), BUV805 conjugated	BD Bioscience	Cat# 748370
Rat anti-mouse CD4 (clone RM4-5), BUV496 conjugated	BD Bioscience	Cat# 741050
Rat anti-mouse CD8a (clone 53-6.7), BUV395 conjugated	BD Bioscience	Cat# 565968
Rat anti-mouse CD8a (clone 53-6.7), BUV615 conjugated	BD Bioscience	Cat# 613004
Rat anti-mouse CD8a (clone 53-6.7), PerCP-Cy5.5 conjugated	Thermo Fisher Scientific	Cat# 45-0081-82
Rat anti-mouse CD25 (PC61.5), PE conjugated	Thermo Fisher Scientific	Cat# 12-0251-81
Rat anti-mouse FoxP3 (clone FJK-16s), FITC conjugated	Thermo Fisher Scientific	Cat# 11-5773-82
Rat anti-human/mouse CD44 (clone IM7), redFluor710 conjugated	Tonbo	Cat# 80-0441-U100
Rat anti-mouse CD279 (clone 29F.1A12), PE-Cy7 conjugated	Biolegend	Cat# 109110
Rat anti-mouse CD279 (clone 29F.1A12), PE conjugated	Biolegend	Cat# 135206
Armenian Hamster anti-mouse CD69 (clone H1.2F3), BB700 conjugated	BD Bioscience	Cat# 566500
Armenian Hamster anti-mouse TCR β (clone H57-597), BV785 conjugated	Biolegend	Cat# 109249
Mouse anti-mouse CD161 (clone PK136), BV605 conjugated	BioLegend	Cat# 108740
Mouse anti-mouse CD161 (clone PK136), PE-CF594 conjugated	BD Bioscience	Cat# 562864
Mouse anti-mouse CD161 (clone PK136), biotin conjugated	Thermo Fisher Scientific	Cat# 13-5941-85
Rat anti-mouse TER-119 (clone TER-119), biotin conjugated	Thermo Fisher Scientific	Cat# 13-5921-82
Rat anti-mouse anti-CD16/CD32 (clone 2.4G2), unconjugated (Fc Block)	Bioceros	N/A

Rat anti-mouse CD16/CD32 (clone 93), unconjugated (TruStain FcX)	BioLegend	Cat# 101320
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Table S3. Overview of all primer sequences used for RT-qPCR

<i>Abca1</i> (Forward) 5'CGTTTCCGGAAGTGCCTA'3
<i>Abca1</i> (Reverse) 5'GCTAGAGATGACAAGGAGGATGGA'3
<i>Abcg1</i> (Forward) 5'TCACCCAGTTCTGCATCCTCT'3
<i>Abcg1</i> (Reverse) 5'GCAGATGTGTCAGGACCGAGT'3
<i>ActB</i> (Forward) 5'GCTTCTAGGCGGACTGTACTGA'3
<i>ActB</i> (Reverse) 5'GCCATGCCAATGTTGTCTCTTAT'3
<i>ApoE</i> (Forward) 5'CTGACAGGATGCCTAGCCG'3
<i>ApoE</i> (Reverse) 5'CGCAGGTAATCCCAGAAGC'3
<i>Apol10b</i> (Forward) 5'TACGACCACCGTTGTGGAAG'3
<i>Apol10b</i> (Reverse) 5'GTGATCCCTGAGGATTTTGAAGT'3
<i>Apol7c</i> (Forward) 5'GAAGCTGCCTTGCAGAGAGT'3
<i>Apol7c</i> (Reverse) 5'TTGTGCCACAGGTCATTTTGG'3
<i>Ccl22</i> (Forward) 5'TCTTGCTGTGGCAATTCAGA'3
<i>Ccl22</i> (Reverse) 5'GAGGGTGACGGATGTAGTCC '3
<i>Ccr7</i> (Forward) 5'TGTACGAGTCGGTGTGCTTC'3
<i>Ccr7</i> (Reverse) 5'GGTAGGTATCCGTCATGGTCTTG'3
<i>Cd40</i> (Forward) 5'TTGTTGACAGCGGTCCATCTA'3
<i>Cd40</i> (Reverse) 5'GCCATCGTGGAGGTACTGTTT'3
<i>Cd80</i> (Forward) 5'GCAGGATACCACTCCTCAA'3
<i>Cd80</i> (Reverse) 5'AAAGACGAATCAGCAGCACAA'3
<i>Cd86</i> (Forward) 5'TGTTTCCGTGGAGACGCAAG'3
<i>Cd86</i> (Reverse) 5'TTGAGCCTTTGTAAATGGGCA'3
<i>Dnase1/3</i> (Forward) 5'CTCTGCTCCTCAATGTGAGG'3
<i>Dnase1/3</i> (Reverse) 5'GCTGCTGTCCTTGATTTCCAT'3
<i>Eno3</i> (Forward) 5'CACAGCCAAGGGTCGATTCC'3
<i>Eno3</i> (Reverse) 5'CCCAGGTATCGTGCTTTGTCT'3
<i>Fdps</i> (Forward) 5'GGAGGTCCTAGAGTACAATGCC'3
<i>Fdps</i> (Reverse) 5' AAGCCTGGAGCAGTTCTACAC'3
<i>Fscn1</i> (Forward) 5'GACTGCGAAGGTCGCTACC'3
<i>Fscn1</i> (Reverse) 5'CTGATCGGTCTCTTCATCCTGA'3
<i>Gapdh</i> (Forward) 5'TGGCAACAATCTCCACTTTGC'3
<i>Gapdh</i> (Reverse) 5'ACAAAATGGTGAAGGTCGGTG'3
<i>Icosl</i> (Forward) 5'TAAAGTGTCCCTGTTTTGTGTCC'3
<i>Icosl</i> (Reverse) 5'ATTGCACCGACTTCAGTCTCT'3
<i>lfi47</i> (Forward) 5'TCTCCAGAAACCCTCACTGGT'3
<i>lfi47</i> (Reverse) 5'TCAGCGGATTCATCTGCTTCG'3
<i>lfit1</i> (Forward) 5'CTGAGATGTCACTTCACATGGAA'3
<i>lfit1</i> (Reverse) 5'GTGCATCCCCAATGGGTTCT'3
<i>lfit2</i> (Forward) 5'AGTACAACGAGTAAGGAGTCACT'3
<i>lfit2</i> (Reverse) 5'AGGCCAGTATGTTGCACATGG'3
<i>lrf7</i> (Forward) 5'GAGACTGGCTATTGGGGGAG'3
<i>lrf7</i> (Reverse) 5'GACCGAAATGCTTCCAGGG'3
<i>lsg20</i> (Forward) 5'GAACATCCAGAACAACCTGGCG'3
<i>lsg20</i> (Reverse) 5'GTAGAGCTCCATTGTGGCCCT'3
<i>Ldlr</i> (Forward) 5'TGACTCAGACGAACAAGGCTG'3
<i>Ldlr</i> (Reverse) 5'ATCTAGGCAATCTCGGTCTCC'3
<i>Nr1h2</i> (Forward) 5'GCCTGGGAATGGTTCTCCTC'3
<i>Nr1h2</i> (Reverse) 5'AGATGACCACGATGTAGGCAG'3
<i>Nr1h3</i> (Forward) 5'CAAGGGAGCAGCTATGTCTG'3
<i>Nr1h3</i> (Reverse) 5'GGACACCGAAGTGCGTTGAG'3

<i>Oas3</i> (Forward) 5'TCTGGGGTCGCTAAACATCAC'3
<i>Oas3</i> (Reverse) 5'GATGACGAGTTCGACATCGGT'3
<i>Rpl13a</i> (Forward) 5'CAGAAAGTTTGCTTACCTGGGG'3
<i>Rpl13a</i> (Reverse) 5'CCGATAGTGCATCTTGGCCT'3
<i>Rac1</i> (Forward) 5'GAGACGGAGCTGTTGGTAAAA'3
<i>Rac1</i> (Reverse) 5'ATAGGCCAGATTCACTGGTT'3
<i>Rac2</i> (Forward) 5'GACAGTAAGCCGGTGAACCTG'3
<i>Rac2</i> (Reverse) 5'CTGACTAGCGAGAAGCAGATG'3
<i>Scube3</i> (Forward) 5'ACGACGGATTTCACCTGGC'3
<i>Scube3</i> (Reverse) 5'GTAGCTGCCCATCATGTTGAC'3
<i>Sdha</i> (Forward) 5'AACTACAAGGGACAGGTGCTG'3
<i>Sdha</i> (Reverse) 5'CTCCCCACAGGCATACAGAC'3
<i>Slco5a1</i> (Forward) 5'ACTTACTAGGTGGGCTACTCATT'3
<i>Slco5a1</i> (Reverse) 5'TTTCCGATGAAACGAGGGTCA'3
<i>Sqle</i> (Forward) 5'ATAAGAAATGCGGGGATGTCAC'3
<i>Sqle</i> (Reverse) 5'ATATCCGAGAAGGCAGCGAAC'3
<i>Srebf2</i> (Forward) 5'GCAGCAACGGGACCATTCT'3
<i>Srebf2</i> (Reverse) 5'CCCCATGACTAAGTCCTTCAACT'3
<i>Tmem176a</i> (Forward) 5'GCCGGATGCTCATTGCTAAG'3
<i>Tmem176a</i> (Reverse) 5'ATGGCCTATGTAGAGGGTTCC'3
<i>Tmem176b</i> (Forward) 5'CCAGTCCGCTCACATCAGC'3
<i>Tmem176b</i> (Reverse) 5'CTGGGTCACCCCTAGAGACAG'3

Supplementary data files (*provided separately*)

Data file S1: Bulk DE genes Homeostatic Mature vs Immature cDC1s.

List of differentially expressed genes between Homeostatic Mature and Immature cDC1s as determined by edgeR differential expression analysis. Benjamini-Hochberg correction was used to adjust the p-values for multiple testing. The listed DE genes have an adjusted p-value smaller than 0.05 and a log2-ratio > 1 or < - 1.

Data file S2: CITE-seq antibody panel. Mouse cell surface protein antibody panel containing 160 oligo-conjugated antibodies (TotalSeq-A, BioLegend) and 9 TotalSeq-A isotype controls.

Data file S3: SCT markers Annotated clusters CITE-seq. RNA markers for the annotated clusters in the CITE-seq experiment. Marker genes were determined by differential expression analysis for each cluster versus all other clusters. This was performed by running the Seurat function FindAllMarkers with default parameters, which utilizes the Wilcoxon Rank Sum test. Markers are ordered according to the appended score column (score = "pct.1/(pct.2+0.01)*avg_logFC").

Data file S4. ADT markers Annotated clusters CITE-seq. Surface protein markers for the annotated clusters in the CITE-seq experiment. Surface protein markers were determined by differential expression analysis for each cluster versus all other clusters. This was performed by running the Seurat function FindAllMarkers with default parameters, which utilizes the Wilcoxon Rank Sum test. Markers are ordered according to the appended score column (score = "pct.1/(pct.2+0.01)*avg_logFC").

Data file S5. ADT and SCT transition markers CITE-seq. RNA and surface protein markers which are differentially expressed between consecutive cDC1 maturation stages in the CITE-seq experiment. These transitional marker genes and surface protein markers were determined by differential expression analysis between consecutive cDC1 maturation clusters. This was performed by running the Seurat function FindMarkers, which utilizes the Wilcoxon Rank Sum test, with a logfc.threshold of 0.20 and the results were subsequently filtered by a p_val_adj cut-off of 0.01. Markers are ordered according to the appended score column (score = "pct.1/(pct.2+0.01)*avg_logFC").

Data file S6. Raw data.